Superconductivity: Resistive Transitions and Critical-Field Trends

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We characterized superconducting transitions and critical-field behavior in ten samples spanning elemental metals, alloys, and non-superconducting materials. All samples were cooled in a liquid-helium cryostat and studied by sweeping an applied DC magnetic field while applying a small AC perturbation field. The induced pickup-coil voltage measured by the lock-in amplifier during the DC field sweep was used to identify the critical field(s) and classify the superconducting samples as Type I or Type II, while two control samples exhibited no superconducting transition. In addition, we performed a four-probe resistivity measurement on $SnPb$ to estimate its critical temperature from the onset of the resistive transition. We compared the extracted critical fields with reference values from superconductivity literature.

I. INTRODUCTION

We investigated two properties of superconductors, tied to both the critical temperature and the critical field(s). The first property was the drop in our sample's resistance to about 0Ω, which occurs at its critical temperature, with no external applied magnetic field. This is the point where our sample becomes superconducting. The second property was the Meissner effect. The Meissner effect is the expulsion of magnetic fields from a superconductor in the presence of small external magnetic fields.

To measure resistivity, we sent a current at a known frequency through our sample and measured the voltage drop across our sample. We then converted our current and voltage drop to resistance using Ohm's law. We did this while cooling down our sample in a cryostat, producing a graph of resistivity by temperature. As the sample cooled, resistivity dropped fairly linearly across most temperatures. We observed a sharp drop in resistivity to 0Ω at a critical temperature. We observed this same drop on repeated trials.

To explore the Meissner effect, we swept a DC magnetic field over our samples at a set range at a fixed temperature while adding a small AC field as a perturbation. We tracked how their diamagnetic response changed with the field amplitude. We measured the induced voltage in a solenoid wrapped around each metal to determine the critical field at which our sample's perfect diamagnetic response was destroyed. We graphed our critical fields by temperature for each sample and distinguished between Type I and Type II superconducting behavior.

A Type I superconductor expels magnetic flux below its critical temperature and critical field, remaining in a Meissner state until it reaches its critical field. A Type II superconductor, on the other hand, is characterized by its two critical fields H_{c1} and H_{c2}. Below H_{c1}, the superconductor expels all magnetic flux, but for fields between H_{c1} and H_{c2}, the field begins to penetrate the superconductor in the form of quantized vortices. Above H_{c2} superconductivity is destroyed, and the sample goes into the normal state. The capacity to sustain a superconductive state under stronger fields makes Type II superconductors especially important in modern applications such as MRI systems, particle accelerators, and even quantum computing [7]. More broadly, superconducting materials are relevant wherever highly efficient current transport or sensitive electrical behavior is required.

II. EXPERIMENTAL METHODS

In this section, we describe the experimental apparatus and data-taking procedures used to measure two key superconducting properties: the drop in a sample's resistance to approximately 0Ω at its critical temperature, and the Meissner effect, which we use to determine the critical field(s) and distinguish between Type I and Type II superconductors.

First, we measured the resistance of $SnPb$ using a four-probe configuration to determine the sample's critical temperature. The function generator (A4) supplied an AC drive current through the outer current leads, and the lock-in amplifier (A9) measured the differential voltage drop across the inner voltage leads. Using our known current of 54.8 μA, we computed the resistance as $R = V/I$, where V is the measured four-probe voltage drop and I is the applied current. Voltage and current readings were verified using a digital multimeter (A6).

Second, we measured the samples' critical field(s) to probe the Meissner effect. We cooled a cryostat (A1) containing all ten samples using liquid helium, with liquid nitrogen used for pre-cooling. After cooling, we swept the applied magnetic field using a large DC magnet while applying a small AC modulation field. The induced response was measured with a lock-in amplifier and recorded on a computer during the sweep.

The samples we tested include elemental superconductors, alloy superconductors, and non-superconducting materials.

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Metals	Alloys	Non-superconducting
Pb_2, Pb_1, Pb_0	$SnPb$	Cu
Sn	In_2Bi	$PTFE$
In	In_4Bi	

TABLE I. The ten different samples we tested in this experiment.

A. Temperature Control

During transfer, the liquid helium level was monitored using an impedance (capacitance) meter (A7). As a representative example, a typical liquid-helium transfer increased the day's starting capacitance reading by approximately 1 pF.

For temperatures above 4 K, the cryostat was operated in flowing helium-vapor (gas) mode, with the needle valve regulating helium flow and heater power adjusted to stabilize the sample temperature (A12). Temperature was monitored with a diode thermometer mounted near the sample and read by the temperature controller.

For temperatures below 4 K we filled the cryostat B (A15) chamber with liquid helium, which is at 4.2 K at atmospheric pressure, and switched to pressure-based thermometry. The temperature was determined using the pressure gauge together with a pressure–temperature conversion chart (A13). After filling the B chamber, returning the system to temperatures above 4 K required over 12 hours, so we separated the measurements into above-4 K and below-4 K runs for all samples.

At each target temperature, we performed a magnetic-field sweep to collect data. The chart bellow summarizes the temperature points measured; each temperature point corresponds to one extracted critical field value for that sample.

B. Field Sweep

To determine the critical field(s) of each sample, we performed a DC magnetic-field sweep at a fixed temperature while applying a small AC magnetic field perturbation. The DC and AC magnetic fields were controlled using the magnet control system shown in (A3). Once superconductivity was broken, a pickup coil wound around the sample would acquire an induced voltage from the time-varying magnetic flux due to the AC magnetic field. The DC magnetic field swept slowly enough to only add negligible additional flux inside the solenoid. The magnitude of the applied magnetic field was monitored using a gaussmeter placed inside the magnet assembly (A5). The lock-in amplifier measured this induced voltage at the perturbation frequency of the AC magnetic field during the sweep.

Each sample required a different DC sweep range to capture its critical field feature(s). In addition, the AC-field drive settings depended on the sample and temperature regime. A chart summarizing the DC sweep ranges and the corresponding AC amplitude/offset settings for the waveform generator is included below (A11).

Temp	PB2	PB1	PB0	SN	CU	IN	SNPB	PTFE	IN2BI	IN4BI
1.5	X			X		X	X		X	X
2.0										
2.6				X		X			X	X
3.0	X			X		X	X		X	X
3.2										
3.4				X		X			X	X
3.6	X			X			X		X	X
3.8	X						X		X	X
4.0	X						X		X	X
4.45	X	X	X	X	X	X	X	X	X	X
5.0	X						X		X	X
5.5										
6.0	X						X			
6.5	X						X			
6.6							X			
6.7							X			
6.8	X						X			
6.9	X						X			
7.0	X						X			
7.1	X						X			
7.2	X						X			
7.3	X									
7.4	X									
7.6										
7.8										
8.0	X						X			
T_c (K)	7.2	7.2	7.2	3.72	N/A	3.4	6.85	N/A	3–4	4.1

FIG. 1. Temperatures measured for each sample (X indicates a measurement was taken). The bottom row lists approximate critical temperatures.

Temp	Pb_2	Pb_1	Pb_0	Sn	Cu
Above 4K	Amp 4 offset 2	Amp 4 offset 2	Amp 4 offset 2	Amp 4 offset 2	Amp 4 offset 2
Below 4K	Amp 4 offset 2	Amp 4 offset 2	Amp 4 offset 2	Amp 4 offset 2	Amp 4 offset 2

Temp	In	$SnPb$	$PTFE$	In_2Bi	In_4Bi
Above 4K	Amp 4 offset 2	Amp 6 offset 2	Amp 4 offset 2	Amp 4 offset 2	Amp 4 offset 2
Below 4K	Amp 4 offset 2	Amp 8 offset 4	Amp 4 offset 2	Amp 4 offset 2	Amp 5 offset 2.5

TABLE II. Amplitude and offset settings for DC sweep measurements above and below 4K for all samples.

For each run, we recorded the DC field and lock-in output during the sweep and used these data to extract the critical field(s) at the set temperature. All instrument settings, sweep ranges, and conversion charts needed to reproduce the procedure are provided in the Appendix. A representative example of the recorded sweep data on the computer is shown in (A8).

III. DATA ANALYSIS

Our first experimental goal was to produce zero resistivity when a current is driven through a sample of metal

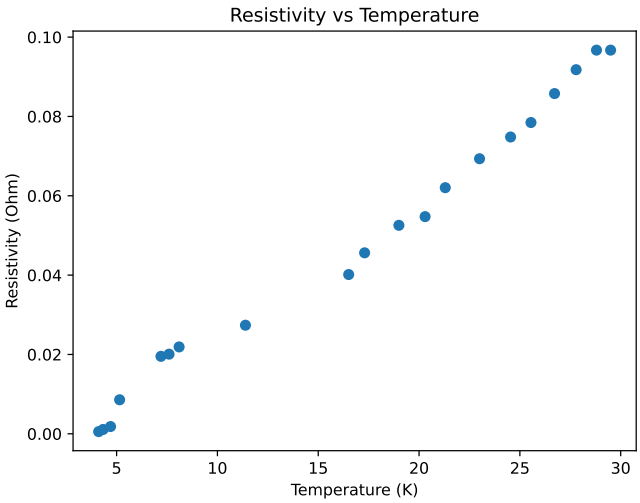

FIG. 2. Resistivity of the SnPb alloy as a function of temperature. The panel shows a large temperature range (4–30 K), with infrequent data collection. The data in the panel lives primarily in the normal regime, with only the left-most data points living in the superconducting regime. We can see a sharp drop in resistivity to near 0 Ω between 5 K and 7.5 K, but cannot observe any further.

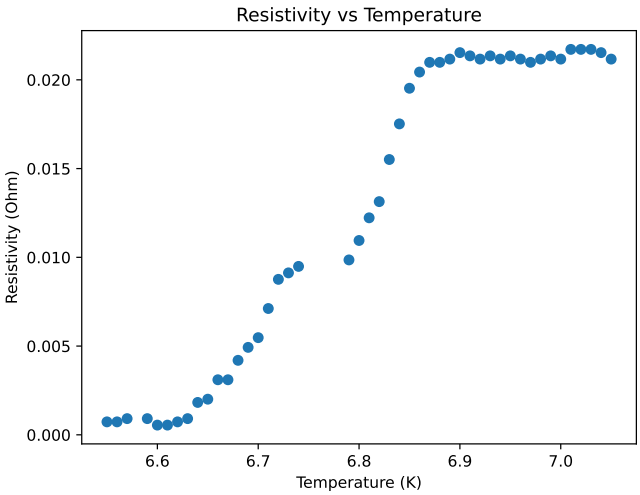

FIG. 3. Resistivity of the SnPb alloy as a function of temperature. The panel shows a small temperature range (6.5–7.1 K), with frequent data collection. The data in the panel lives in a mix of the normal regime, the superconducting regime, and the transition point. Theoretically, this transition is instantaneous, but we observe only a sharp drop in reality. We can see this drop in resistivity to near 0 Ω between 6.6 K and 6.9 K, with a knee visible around 6.85 K.

by cooling it below a critical temperature. Our second experimental goal was to produce superconductivity in various samples of metal by cooling them to temperatures below their critical temperature and then destroying superconductivity by applying a strong enough external magnetic field. Our data collection methods are mentioned in the previous section.

We will now review the data gathered from our resistivity experiments. The range of temperatures observed varied widely across our trials, from 4 K to 100 K. We had two distinct methods of data collection. First, we collected data infrequently across a large range (Fig. 2). We then collected data frequently across a small range near our believed critical temperature (Fig. 3).

Our analytic methodology for the resistivity experiment was purely observational. The crude-grain resistivity graphs were only used to inform the temperature range we chose to focus on for our fine-grain graphs. Our data was first pulled from a video recording of the readings on our temperature controller and our lock-in amplifier (voltage drop), then input into an Excel spreadsheet. Initial voltage vs temperature graphs were generated on Excel before we exported the data into .csv files on VSCode. Using our known current of 54.8 μA, we converted our voltage drops into resistances. We then plotted a resistance vs temperature graph for each trial. We chose not to combine trials as we measured a different set of points for each trial. As mentioned in Fig 3, our critical temperature is between 6.6 K and 6.9 K.

We made several experimental decisions that a different group may have chosen an alternative approach for. We chose not to repeat trials over identical temperature ranges. It would have been perfectly reasonable to repeat data collection across the same temperature range, 3-7 times. This would have led to greater confidence in the regime transition points and what the transition looked like. We chose not to because we felt it was important to progress to studying the Meissner effect (which took much longer), and due to our difficulties in lowering our sample temperature at a precise rate. As mentioned above, our temperature was lowered through a combination of our controller's heater and by pumping liquid helium through the capillary in the cryostat. It proved very difficult to do this precisely and at the level of detail we wanted to, so we stopped after we achieved it once.

Our other major decision was to report our observed critical temperature purely through visual inspection. Because of this, we left our critical temperature as being within a range we were fairly confident in. We chose not to perform any in-depth analysis due to our lack of available data.

We will next review the data gathered from our Meissner experiments. Our intention with this data was to produce a Meissner graph of critical field vs temperature for each sample, as well as a predictive model. This was intended to show how superconductivity is dependent on both sample temperature and external magnetic field. We recorded data on 10 different samples (a mix of elemental metals and alloys) across 24 different temperatures, with our external DC magnetic fields swept to a maximum amplitude between 0.25 kG and 1.4 kG. All magnetic field sweeps swept across 0 applied external field. This data was exported to VSCode in .txt files. We encountered three types of noise potentially detrimental

to our critical field analysis. One example of our raw data and a cleaned version can be seen for a Type I and a Type II superconductor in Fig 4 and Fig 5 below.

The first noise issue we dealt with was an initial spike in our external DC field. This caused a huge flux to flow through our solenoid, causing a huge spike in induced voltage. As mentioned in our Methods section above, under expected conditions, only the AC field would cause magnetic flux through the solenoid. The amplitude of the AC field was significantly smaller than the amplitude of the DC field, so this initial, large, induced voltage was significantly larger than any other induced voltage we encountered. Further, as this was instantaneous, we could not estimate the critical field(s) of our samples from this initial spike. As it was well outside of expected trial conditions, we decided that we could reasonably remove it.

To do so, we first checked the time lock-in constant that we had recorded from our lock-in amplifier(A9). Unfortunately, we were not diligent in recording this value. For most of our data, we found that it was 300 µs, but for a few trials, it was 100 µs. On one day, we did not record our time lock-in constant. As we couldn't be exactly certain, after looking through all of our data, we chose 300 µs as our lock-in constant. We first made sure we didn't lose any important data with this choice. The error due to this lock-in is expected to be about $\frac{1}{e^5} \approx 0.7\%$ of our initial error after 5 times the length of our lock-in constant. After this point, the error is negligible and unlikely to cause any abnormalities in our analysis

Our second noise issue was due to accidental brushes with experimental equipment, primarily our BNC cables. Brushing these caused large spikes in our induced voltages at the time of collision, but no spike in the applied magnetic field. These spikes were typically around 10 to 20 data points wide, within a data set of around 4800 points, typically. Leaving these in rendered some of our graphs difficult to look at visually. They also complicated our analysis. There is a small noise spike of this type in the top central graph in Fig 4.

To combat this, we introduced a median filter. This filter replaced each data point with the median of its 13 nearest neighbors. We chose 13 after considering other numbers, as the best value to balance noise reduction and information loss in our data. This completely removed all noise spikes from our data, improving both our visualized data and our later analysis.

The last noise issue was due to random measurement noise between consecutive data points. While this flutter was very small and unobstructive visually, the time distance between consecutive points was also very small. Our first and second derivatives at these points of our induced voltage with respect to time were huge, which was detrimental to our analysis methods.

We combated this differently for Type I and Type II superconductors. We considered any elemental metal to be a Type I superconductor and any alloy to be a Type II superconductor. While this is true here and generally true, it is important to note that this is not always true. Niobium, an elemental metal, is a Type II superconductor. Tantalum disilicide, an alloy, is a Type I superconductor[6]. For our Type I superconductors, we used a Gaussian filter to smooth out the noise in our data, reducing the tiny jumps from consecutive points. This did not work for our Type II superconductors, as it smoothed our data too much. For them, we used a Savitzky-Golay to smooth our data. The Savitzky-Golay filter performs localized polynomial regression on our data set. We set our local window to 201 data points, which we felt worked best after adjusting it. We also chose the end behavior to mirror the local behavior, so as not to have our analysis biased by these edge cases.

Before producing our Meissner graphs, we produced 3 graphs for each sample at each temperature we recorded at. The first chart was *DC Field Strength vs Time*. The second was *Induced Voltage vs Time*. The last was *Induced Voltage vs DC Field Strength*. Though the polarity of our field was negative, we flipped it for our graphs, as we only cared about the absolute strength of the magnetic field. The first two graphs were plotted on scatter plots, using the cleaned data. The last plot was made by first splitting our field strength array into an upward sweep and a downward sweep. We then split our time and induced voltage arrays in two, using the same turning point we did for our field strength array. The split time arrays are not used here, but are used later to calculate first and second derivatives in our analysis. Our last plot simply plots the two sweeps of the DC field against their respective induced voltages. We used our last plot, *Induced Voltage vs DC Field Strength*, to do quality control for our Meissner graphs, later in our analysis.

Type I Superconductor (Pb_2, $T = 3.97\,\mathrm{K}$)

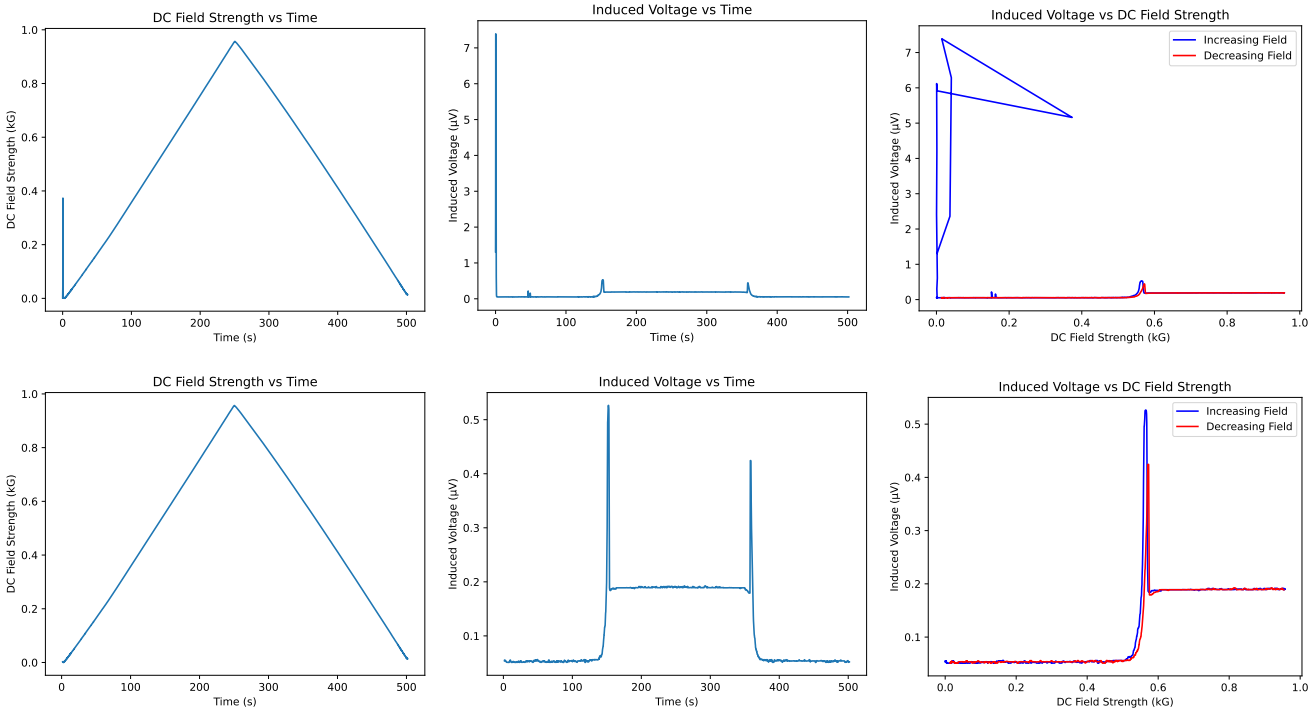

FIG. 4. Comparison of raw and filtered measurements for a Type I Pb_2 sample at $T = 3.97\,\mathrm{K}$. The top row shows the unfiltered experimental signals: the DC magnetic field sweep, the induced voltage measured by the lock-in amplifier, and the resulting $\frac{dM}{dH}$ signal. The bottom row shows the same data after applying the filtering procedure used in our analysis.

Type II Superconductor (SnPb, $T = 3.8\,\mathrm{K}$)

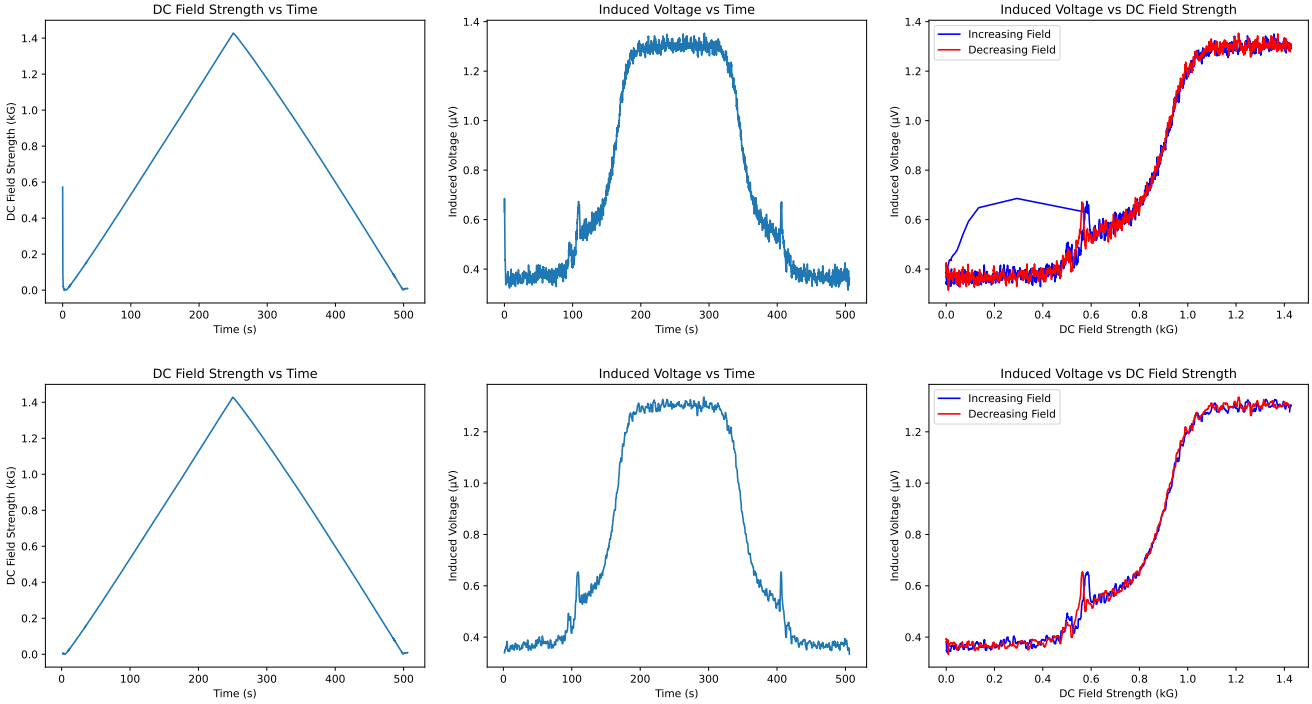

FIG. 5. Comparison of raw and filtered measurements for a Type II SnPb sample at $T = 3.8\,\mathrm{K}$. The top row shows the unfiltered data, while the bottom row shows the filtered signals used to extract the superconducting critical field.

According to theory, a Type I superconductor at the critical field resembles a step function, where the diamagnetism jumps for perfectly opposed to the external magnetic field to no diamagnetism. For a step function, its derivative is a delta function. Experimentally, we see something very similar. Our elemental metals resemble a step function, with what looks like Gibbs phenomena at the critical temperature. Note that we are not suggesting that this is Gibbs phenomena, merely that it looks like it. Similarly, its derivative is an approximation of a delta function, with a broadened spike.

We decided to use the location of the peak of this spike as our critical field, because according to the theory, this is where it should be. Another group may have reasonably chosen the beginning of this spike or the left side of the full-width at half maximum (FWHM) as the location of the critical field. To find these peaks, we computed the first derivative of our induced voltage with respect to time. We then used NumPy's *max* and *min* functions to find the maximum of our derivative on our upward sweep and the minimum of our derivative on our downward sweep. We then found the DC magnetic field associated with these two values, averaged them, and provided that as our critical field. We used the minimum derivative for the downward sweep because the metal is regaining superconductivity, the step function is in the opposite direction, and so is the delta function.

Type II superconductors are different. The diamagnetism in a Type II superconductor is not destroyed all at once when the critical field is reached. Instead, Type II superconductors have two critical fields. Prior to reaching the first critical field, the metal is a perfect diamagnet. Between the two fields, the metal is an imperfect diamagnet, no longer perfectly opposing the external magnetic fields. Once the second critical field is reached, the metal completely loses its diamagnetism. In theory, Type II superconductor has two knees, one at the first critical field and one at the second critical field. Our experimental data matched this fairly closely.

To find these knees, we took the second derivative of our induced voltage with respect to time. Mathematically, because our induced voltage is roughly constant outside of the mixed regime, our two knees will correspond to the positive and negative spikes that we see in our second derivative. Again, we used NumPy's *max* and *min* functions. Contrary to above, for the first critical field, we took the average of the maximum second derivative from our two sweep directions. For the second critical field, we took the average of the minimum of the second derivative from our two sweep directions.

Reasonably, a group may have chosen to obtain these critical fields visually, as opposed to using derivatives like we did. Even with our data cleaning, noise was a large issue. For some of our graphs, we needed to manually input the correct (approximate) critical fields, through visual observation. This is discussed more in-depth below. However, we felt that this would lead to reproducibility issues. This method is difficult to improve due to the subjectivity inherent in the analysis. Another group may also have decided not to take the critical field as an average of the upward and downward sweeps. We did so to combat incidental hysteresis effects and any additional error caused by the experimental geometry.

For each sample metal, we stored the array(s) of critical fields along with a corresponding array of temperatures. This became our data, which we used to create our Meissner plots.

We chose to use a theory-based model to conduct regression analysis on our data. We opted to use the theoretical model instead of creating our own model or trying various polynomial and exponential models due to our lack of knowledge in the area. Another idea was to use machine learning to train potential exponential and polynomial models. However, we felt that we didn't have enough data to justify splitting our data into training and test data. Also, building a machine learning model would have added an extra time cost. We felt this was not worth it, given the available theoretical model. The model we chose was:

$$H(T) = H_0 \left[1 - \left(\frac{T}{T_c} \right)^2 \right]$$

We used Python's *curve_fit* function to optimize our model's H_0 and T_c with respect to the data for each sample. For Type II superconductors, this was a little more complex. With two critical fields, optimizing models to each field completely separately would have resulted in distinct T_cs with respect to each field. This was something that we did not want. Using a few tricks, we optimized our model for a joint T_c, while still keeping our critical field data separate. This ensured the model for each field was only dependent on the data from that field. From our model, we pulled the critical temperature and the critical field (at 0 K). We also gave a confidence interval of twice the standard deviation. For a large enough sample size, this would be about a 95% confidence interval. Here, because our amount of data is so small, this is significantly less than a 95% confidence interval. We chose not to use a Student-T distribution to keep our analysis simple. Another group would have very reasonably done so.

For all of our models, we included confidence intervals not only for our parameters, but also for our model itself. We used standard error propagation techniques to produce our standard deviation for H, for every $T \in [0, T_c]$. Our error propagation formula was:

$$\sigma_H^2 = \left(\frac{\partial H}{\partial H_0} \right)^2 \sigma_{H_0}^2 + \left(\frac{\partial H}{\partial T_c} \right)^2 \sigma_{T_c}^2 + 2 \frac{\partial H}{\partial H_0} \frac{\partial H}{\partial T_c} \mathrm{Cov}(H_0, T_c)$$

Similarly as above, we opted to give a confidence interval of twice the standard deviation, instead of computing a 95% confidence interval using a Student-T distribution.

For each sample, we produced a Meissner graph. Each graph contains the sample data, our model & confidence

intervals, and the estimated critical temperature and critical fields (at 0 K) with their confidence intervals. These graphs can be seen in Fig 6.

We then did quality control on each graph, visually inspecting them and comparing the found field values with our *Induced Voltage vs DC Field Strength* graph. We only found an issue with one sample, $SnPb$. At 1.5 K, the data was too noisy for us to feel confident estimating critical fields from it. Therefore we felt it was appropriate to remove the file from our analysis. At 3.0 K, our first calculated critical field was greater than our second calculated critical field, indicating that this file broke our analysis. Instead of reworking our analysis, we visually inspected the *Induced Voltage vs DC Field Strength* graph and produced our first and second critical fields in this manner.

In summary, our analysis found that the critical temperature for our $SnPb$ alloy is somewhere in the range $(6.6K, 6.9K)$. Further experimentation with more precise temperature control is needed to narrow this range and produce a more exact estimate of the critical temperature. It's worth noting here that in our Meissner experiment, our estimated critical temperature for $SnPb$ was 7.29 ± 0.157 K. However, we cannot be certain that the alloy composition is exactly the same for both samples is identical. As Sn has a lower critical temperature and Pb has a higher critical temperature than our alloys, both of these critical temperatures are reasonable.

We found our theoretical model to be a reasonable fit for our data. In general, our derived critical fields and critical temperatures did not precisely match our theoretical critical temperatures and critical fields.

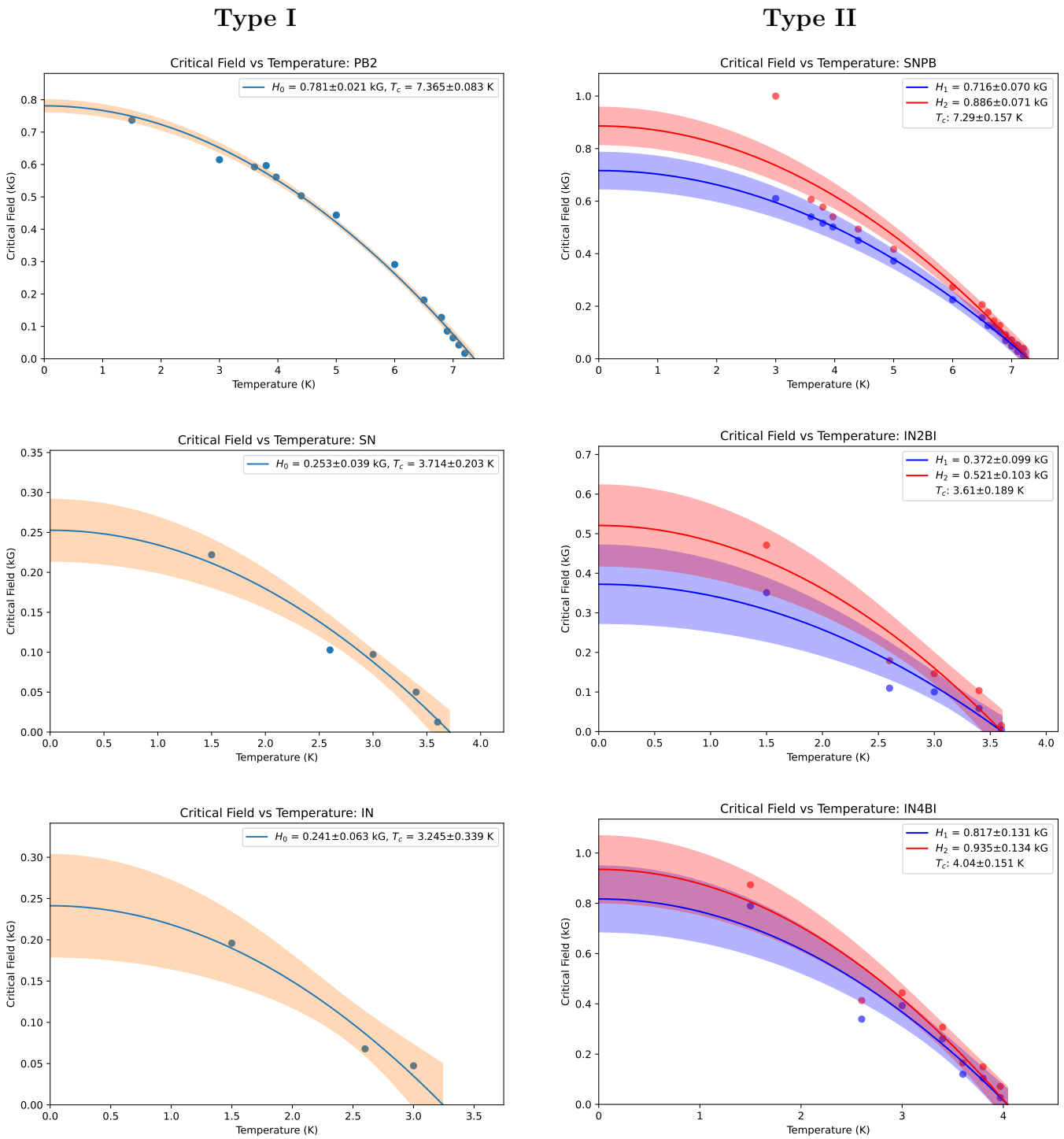

FIG. 6. Critical field as a function of temperature for six samples. Left column: Type I superconductors (Pb_2, Sn, In). Right column: Type II superconductors (SnPb, In_2Bi, In_4Bi). Points show extracted critical field values and curves show best-fit trends with uncertainty bands.

IV. DISCUSSION

As mentioned in our analysis, the critical temperatures and critical fields that we produced did not exactly match our theoretical values. Further, our critical fields at a given temperature did not exactly match previous experimental findings. This is not surprising. We only took one data point for each sample at each temperature. We may have made systematic errors on these trials or encountered outliers. Without more data, we are be hesitant to suggest that the previously derived values should be reconsidered. Instead, we would like to collect more data and further investigate our experimental setup, to better understand how this may affect the data we collect. Investing in a more precise temperature controller, as well as a mechanical pressure pump, would be worthwhile.

Initially, we had set out to also derive the magnetization of our samples as a function of the field and the susceptibility of our samples. We did not do this directly, but it can be seen indirectly in our analysis. Our magnetization is proportional to our induced voltage, which we did measure. In the superconducting regime, our magnetization and susceptibility is 0. Note that there appears to be some induced voltage in our superconducting regime, indicating magnetization. However we believe this to be a byproduct of our set up, not due to our sample's being magnetized. At during the regime transition, our susceptibility becomes non-zero. Once in the normal regime, our magnetization was typically positive and constant and our susceptibility was 0. For a few of our trials, we found increasing magnetization and non-zero susceptibility in the normal regime.

There are several peculiarities in our data that deserve mention. For several of our trials for Pb_2 and Sn, our induced voltage increased during the normal regime. As mentioned above, our external AC field, which induces this voltage, is constant and our external DC field sweeps slow enough as to not induce any voltage in the solenoid. Therefore, this increase in voltage is likely due to the changing magnetization of our sample, due to a non-zero magnetic susceptibility. Our sample appears to be behaving like a paramagnet. This property is likely due to our experimental geometry, possibly due to hysteresis and eddy currents formed in our sample. It would be beneficial to test different sample geometries in the normal regime, where this occurs.

A keen eye may notice that our induced voltage is not always $\sim 0\mu V$ in our *Induced Voltage vs Time* and *Induced Voltage vs DC Field Strength* graphs. Our understanding is that this may have been due to our experimental setup or an accidental offset somewhere in our data. Similarly, while our scale was mostly consistent, there are a few graphs with a different scale. While these errors are frustrating, they appeared to have no real effect on our analysis. Our critical fields still occur where we might expect them too. As such, we chose not to investigate this issue in-depth.

V. REFERENCE

This section summarizes literature values for the critical magnetic fields of the materials studied in this experiment. These reference values provide a benchmark for comparison with the experimental results obtained in our measurements.

Sample	Critical Field(s) at $T \to 0$ (kG)	Ref.
	Type I Superconductors	
Pb_2, Pb_1, Pb_0	$H_c = 0.803 \pm 0.001$	[2]
Sn	$H_c = 0.305 \pm 0.002$	[2]
In	$H_c = 0.2815 \pm 0.002$	[2]
	Type II Superconductors	
SnPb	$H_{c1} = 0.229 \pm 0.007$, $H_{c2} = 0.877 \pm 0.013$	[5]
In_2Bi	$H_{c1} = 0.500$, $H_{c2} = 1.100$	[3]
In_4Bi	$H_{c1} = 0.280$, $H_{c2} = 1.200$	[4]
	Non-Superconducting Reference Samples	
Cu	N/A	[8]
PTFE	N/A	[9]

TABLE III. Zero-temperature critical fields for the samples used in this experiment. Values are converted from Gauss to kG.

VI. CONCLUSION

In this experiment we investigated superconducting behavior in ten different samples placed in a cryostat cooled by liquid helium. First we used a four-probe resistivity measurement for $SnPb$ and revealed a superconducting transition between $6.6\,K$ and $6.9\,K$. Critical fields were studied by sweeping an applied DC magnetic field while applying a small AC perturbation field. From these measurements, we classified Pb_2, Pb_1, Pb_0, Sn, and In as Type I superconductors, each with a single H_c critical field. The alloy samples $SnPb$, In_2Bi, and In_4Bi displayed Type II superconducting behavior with two critical fields H_{c1} and H_{c2}. In contrast, the control samples Cu and $PTFE$ exhibited no superconducting transition. Although some of the extracted critical temperatures and fields differ from literature values, the overall temperature dependence and qualitative behavior are consistent with theoretical expectations for superconductors. Remaining discrepancies are likely due to the limited amount of collected data, the precision of the measurement instruments, and systematic uncertainties associated with the experimental setup.

Appendix A: Appendix: Apparatus Photos (External Links)

A1. Cryostat.

A2. 50 Ohm Resistor.
A3. AC and DC magnet field generator controls.
A4. Waveform generator used for the AC magnetic field.
A5. Gaussmeter inside AC magnets.
A6. Multimeter used for voltage/current verification.
A7. Impedance meter used to monitor helium level.
A8. Example data recorded during a DC field sweep.
A9. Lock-in amplifier.
A10. Pressure gauge used for pressure-based thermometry.
A11. Waveform generator used for the DC magnetic-field sweep.
A12. Temperature controller used to stabilize the sample temperature.
A13. Pressure–temperature conversion chart.
A14. GitHub repository containing the Python scripts used for experimental data processing and analysis.
A15. Cryostat chart.

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